



Department of CSE

Mini Project Report

“ Malware Family Classification Using a CNN with Attention on the MaleBin Dataset ”

Course Code: ICE 478/CSE475

Section: 2

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1. Introduction

Malware detection has become an important research area in cybersecurity because new malware variants are continuously being developed to evade traditional detection systems. In recent years, researchers have applied deep learning techniques to malware classification by converting malware binaries into grayscale or RGB images. These image-based approaches reduce the need for manual feature engineering and allow deep learning models to learn useful patterns directly from malware images.

This report reviews five recent studies related to malware image classification. The objective is to understand the methods used by previous researchers, compare their strengths and limitations, and identify research gaps. The findings from these studies provide useful insights for developing the proposed CNN with an attention mechanism using the MaleBin Malware Binary Greyscale Images dataset.

2. Related Work Summary

Reference	Year	Dataset	Method	Key Findings
Self-Supervised Vision Transformers for Malware Detection	2022	AndroZoo	Vision Transformer	Achieved high detection accuracy using self-supervised learning.
Grayscale Image-Based Deep Learning Model for Efficient and Scalable Malware Detection	2025	MalImg	CNN	Demonstrated effective malware classification using grayscale images.
Multimodal Malware Classification Using Proposed Ensemble Deep Neural Network Framework	2025	Android Malware Dataset	Ensemble Deep Learning	Improved classification by combining multiple feature sources.

YoloMal-XAI: Interpretable Android Malware Classification Using RGB Images and YOLO11	2025	CICMalDroid2020	YOLO11 + XAI	Combined accuracy explainable techniques.	high with AI
Hybrid Noise-Resistant Technique for Malware Classification	2026	BODMAS	CAE + LBP + XGBoost	Achieved accuracy robustness noisy samples.	high and against malware

Table 2.1: Summary of Related Studies

3. Literature Review

3.1 Self-Supervised Vision Transformers for Malware Detection (2022)

This study proposed a malware detection framework based on a Self-Supervised Vision Transformer (ViT). Instead of relying on manually extracted features, malware binaries were converted into grayscale images and used to train the transformer model. The self-supervised learning strategy enabled the model to learn meaningful image representations from a large amount of unlabeled data before fine-tuning for malware classification.

The experiments were conducted using the AndroZoo dataset and produced promising classification results. The authors showed that the self-attention mechanism effectively captured relationships between different regions of malware images, improving feature extraction compared with conventional CNN models.

Although the proposed method achieved high accuracy, it required a large training dataset and significant computational resources. These requirements make Vision Transformers less practical for projects with limited hardware. This study highlights the importance of attention mechanisms, which motivates the use of attention in the proposed CNN model.

3.2 Grayscale Image-Based Deep Learning Model for Efficient and Scalable Malware Detection (2025)

This paper presented a CNN-based framework for malware classification using grayscale images generated from malware binaries. The objective was to develop an efficient end-to-end model

capable of learning discriminative features automatically without requiring handcrafted feature extraction.

The model was evaluated using the MalImg dataset and achieved strong classification performance. The results demonstrated that CNNs can effectively identify malware families by learning texture and structural information directly from grayscale images. The simple architecture also reduced preprocessing complexity.

However, the model did not include any attention mechanism, meaning that all image regions were treated equally during feature extraction. In addition, the experiments were limited to a single dataset. This work provides a useful CNN baseline for the proposed project, which extends the approach by incorporating an attention mechanism to improve feature learning.

3.3 Multimodal Malware Classification Using Proposed Ensemble Deep Neural Network Framework (2025)

This research introduced a multimodal malware classification framework that combined image-based features with additional malware information. An ensemble deep neural network was used to integrate different feature representations and improve classification accuracy.

The study reported promising results, showing that combining multiple feature sources can increase the robustness of malware classification. Ensemble learning also helped improve performance when dealing with complex malware datasets.

Despite its advantages, the framework required multiple preprocessing steps and feature extraction methods, making the overall system more complex. Compared with this approach, the proposed CNN with attention focuses on a simpler end-to-end architecture that can automatically learn features from malware images without relying on multiple feature sources.

3.4 YoloMal-XAI: Interpretable Android Malware Classification Using RGB Images and YOLO11 (2025)

This study proposed an explainable malware classification framework called **YoloMal-XAI**. The authors converted Android malware binaries into RGB images and used the YOLO11 model for classification. To improve model transparency, they applied Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM to highlight the image regions that influenced the model's predictions.

The framework was evaluated using the CICMalDroid2020 dataset and achieved very high classification accuracy. Besides accurate predictions, the explainability methods helped users understand how the model reached its decisions, making the system more reliable.

However, the framework requires RGB image conversion and a computationally expensive model, making it less suitable for environments with limited resources. This study demonstrates the importance of model interpretability, while the proposed work focuses on improving feature learning through an attention mechanism within a simpler CNN architecture.

3.5 Hybrid Noise-Resistant Technique for Malware Classification (2026)

This paper introduced a hybrid malware classification framework that combines a Convolutional Autoencoder (CAE), Local Binary Pattern (LBP), and XGBoost classifier. The objective was to improve malware classification accuracy, especially when malware images contain noise.

The experimental results showed excellent classification performance on the BODMAS dataset. By combining deep learning with handcrafted texture features, the proposed framework achieved robust performance under different conditions.

Although the model performed well, it required additional feature extraction and preprocessing, increasing the complexity of the overall system. In comparison, the proposed project aims to develop a simpler end-to-end CNN model with an attention mechanism that can automatically learn important features from grayscale malware images.

4. Comparative Analysis

Method	Feature Extraction	Complexity	Main Strength	Limitation
Vision Transformer	Automatic (Self-Attention)	High	Learns global image features effectively.	Requires large datasets and high computational resources.
CNN	Automatic	Low	Simple and efficient for malware image classification.	May overlook important image regions without attention.

Ensemble Deep Learning	Multiple Features	Medium	Improves classification by combining different features.	Requires additional preprocessing.
YOLO11 + XAI	Automatic + Explainability	High	High accuracy with interpretable results.	Computationally expensive.
Hybrid (CAE + LBP + XGBoost)	Automatic + Handcrafted	High	Robust against noisy malware images.	More complex implementation.
Proposed CNN + Attention	Automatic + Attention	Medium	Balances accuracy, efficiency, and simplicity.	Performance will be evaluated experimentally.

Short Discussion

The reviewed studies use different approaches for malware classification, each with its own advantages and limitations. Transformer-based and YOLO-based models achieve high accuracy but require greater computational resources. CNN-based methods are simpler and more efficient but can benefit from attention mechanisms. Hybrid and ensemble approaches improve feature representation but increase preprocessing complexity. The proposed CNN with an attention mechanism aims to provide a balance between performance, computational efficiency, and implementation simplicity.

5. Research Gap

The reviewed studies show that deep learning has significantly improved malware image classification. However, several limitations still exist. Many CNN-based models do not use attention mechanisms, which may reduce their ability to focus on important image regions. On the other hand, Vision Transformer models provide strong performance but require large datasets and expensive hardware.

Some studies improve accuracy by combining multiple feature extraction methods, but these approaches increase preprocessing complexity. In addition, only a limited number of recent studies have used the MaleBin Malware Binary Greyscale Images dataset.

To address these limitations, this project proposes a CNN integrated with an attention mechanism. The goal is to automatically learn important image features while keeping the model simple, efficient, and suitable for malware family classification.

6. References

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